



MAK ENGINEERS



FLAME JET MODEL: K6

- All Purpose Ferrous & Non-Ferrous Coating
15G, 11G, 1/8" (3.17mm) 3/16" (4.76mm) WIRE

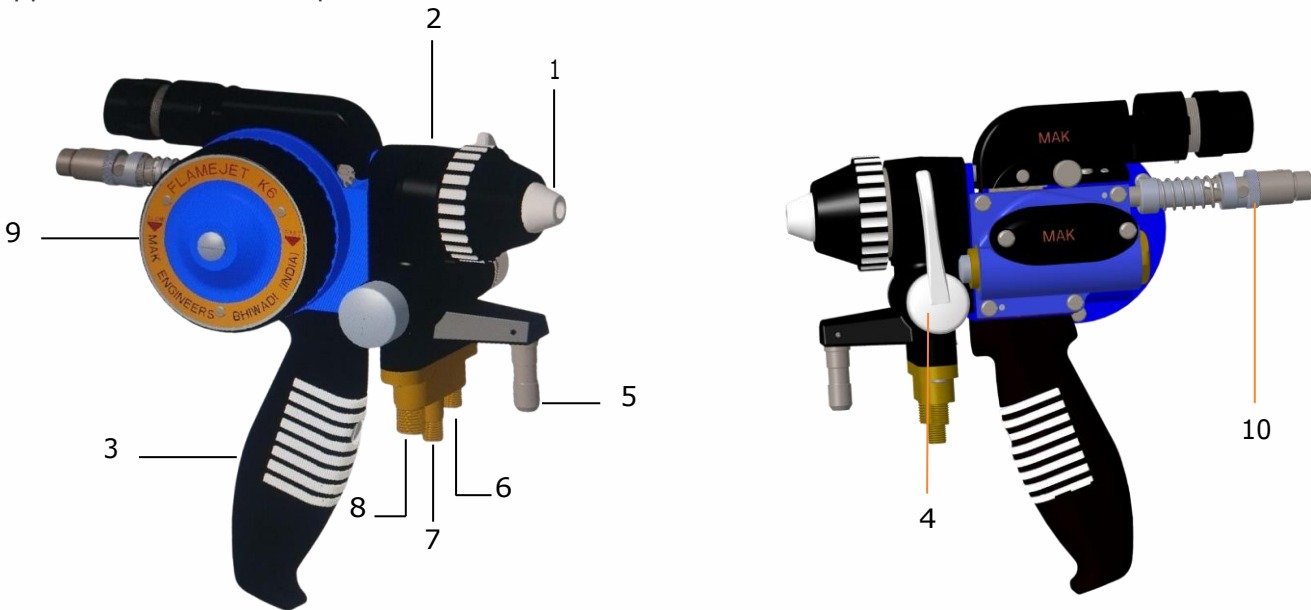
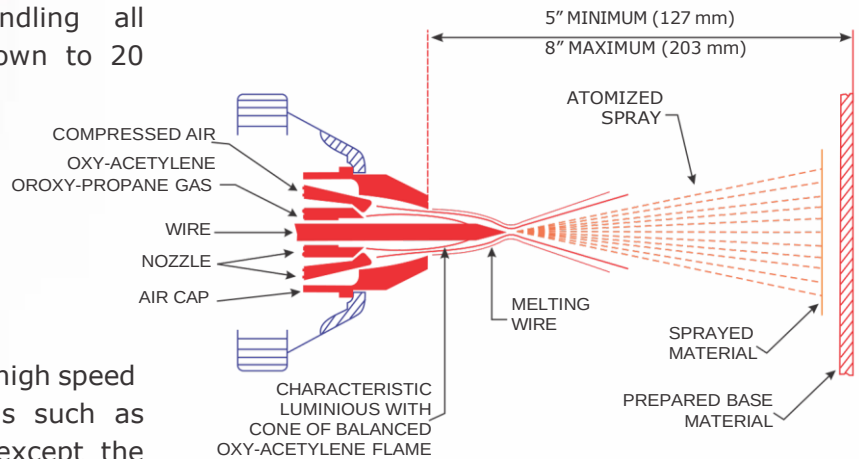
Wire Flame Spray Process

The wire is fed in to the front of gun through air driven turbine. Fuel gas and oxygen is properly and safely mixed in siphon plug gas head assembly which provides combustible heat source. The

molten metal is hence atomized and propelled by compressed air in the form of fine metal spray for coating on job surface.

With universal, forged-aluminum gas head it can be adapted to use any fuel gas. The standard gun is set up to spray 1/8" wire using acetylene as the fuel gas suitable hardware is available for handling all standard wire sizes ranging from 3/16" down to 20 gauges.

The controlled speed range of the Type K6 Gun is from about 1-1/2 feet of wire per minute with standard gears to approximately 45 feet per minute with the special, easily installed, high speed gear set. The high speed gear set is for continuous high speed spraying of the lower melting point metals such as Babbitt, tin and zinc. The entire system (except the compressor), can be mounted on a cart for mobile applications and field repair work.



- | | | |
|---------------------------|--------------------------------------|------------------------------|
| 1. Gas Nozzle and Air Cap | 2. Gas Head Assembly and Siphon Plug | 3. Gun Handle |
| 4. Valve Handle Assembly | 5. Gun Mounting Stud | 6. Fuel Gas Connection |
| 7. Oxygen Connection | 8. Air Connection | 9. Turbine & Wire Drive Unit |
| | 10. Snubber Assembly | |

The size of K6 Flame Spray Gun is approximately 9" Long & 9" height and 5-1/4" Wide. Weight is approximately 2.7 Kg.

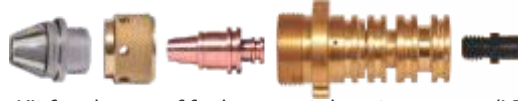
Spray Coating Wires

Generally all metals and metal alloys in wire form are used for spray coating i.e. Zinc, Copper, Aluminium, Brass, Steel, Stainless Steel, Molybdenum, Nickel Alloy etc. Standard Wire used are 1/8" (3.17mm), 5/32" (3.97mm), 3/16" (4.76mm).

Combustion Gases Used

The combustion gases used are oxygen and fuel gas such acetylene, LPG, propane etc.

CONVERSION KIT



Conversion Kit for change of fuel gas acetylene to propane/LPG or vice-versa.

FLAME JET-K6 SPECIFICATIONS

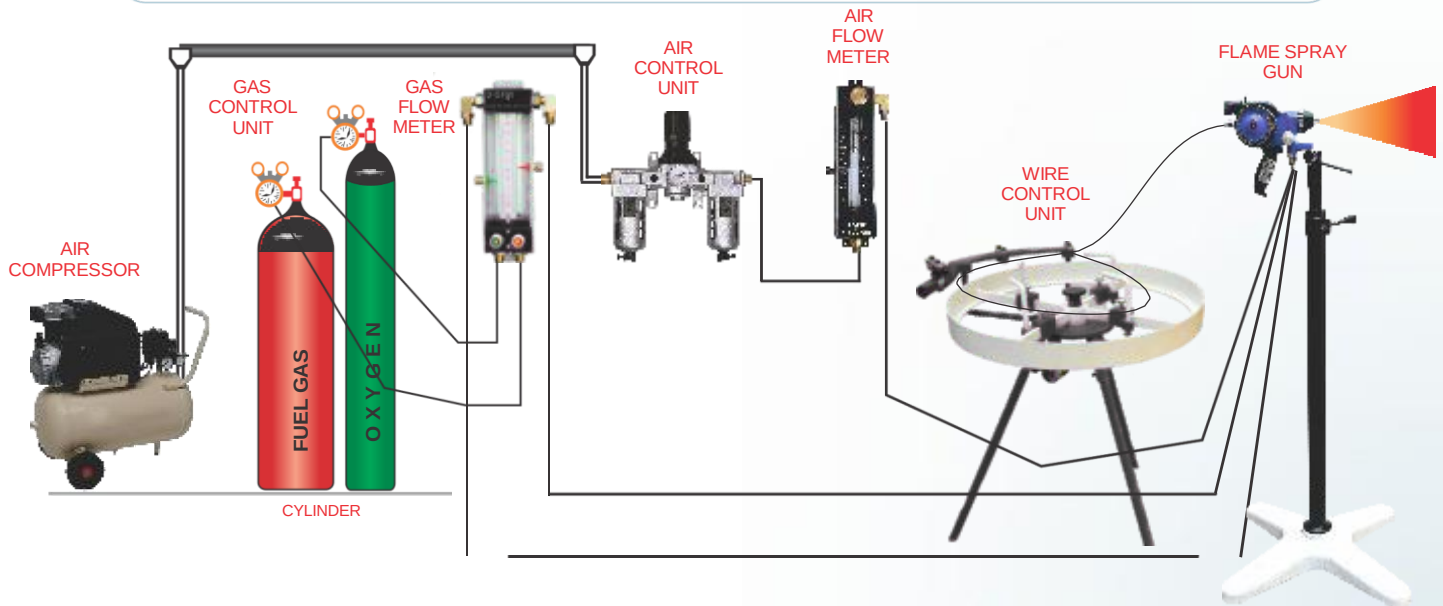
Operation	Manual
Wire Feed	Air Turbine
Wire Size	1/8" (3.17mm)
Standard Wire	15G, 11G, 5/32", 3/16" (4.76mm)
Size Optional	Oxygen and Fuel Gas
Gasses Used Fuel	Acetylene or LPG / Propane etc.
Gas Compressed	35 CFM at 75 psi (5 bar)
Air Net Weight	Approx 2.7 Kg. (Gun Only)

Gun is supplied with standard spares and tool kit.

TYPICAL APPLICATION

- To provide Anti-Corrosion Coating
- To Improve Wear Resistance Properties
- To Reclaim Worn out Parts
- To Enhance Electrical Conductivity
- To Create Thermal Barrier Coating
- To Decorate Articles of various Materials

FLAME JET K6- COMPLETE INSTALLATION



Typical Spray Rate with 1/8" wire (For Guidance Purpose Only)

Aluminium	: 5.0 Kg./Hour
Zinc	: 14.0 Kg./Hour
Carbon Steel	: 5.9 Kg./Hour

Note: Buyer to specify wire diameter, material and fuel gas to be used while ordering the gun.

Copper	: 10.9 Kg./Hour
Nickel	: 4.5 Kg./Hour :
Molybdenum	3.5 Kg./Hour

Note: The above parameters are for reference purpose only. The actual values may differ depending upon the geometry & job size.

OPTIONAL ACCESSORIES

Gas Control Unit

Consist of Gas Regulator with pressure gauge for regulating gas flow to FLAME JET. It has two pressure gauge to indicate cylinder and delivery pressure.

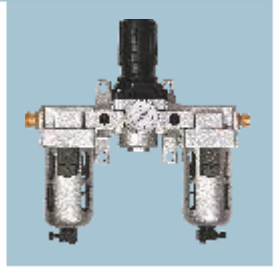
Pressure Rating:	Oxygen	: 10 Bar
	LPG/Propane	: 06 Bar
	Acetylene	: 2.5 Bar



Air Control Unit

Consist of air filter cum regulator for secondary filtration and regulating air flow to FLAME JET. An additional outlet provides connections for operator's helmet.

Pressure Rating: 10 Bar



Hose Kit

Consist of Hose to connect oxygen cylinder & fuel gas cylinder with gas flow meter to spray gun.

Oxygen Hose	- Blue	: 3+7 Mtrs.
Fuel Gas Hose	- Red	: 3+7 Mtrs.
Air Hose	- Black	: 7 Mtrs.



Air Flow Meter with Hose

The 3AFM Air Flow Meter is a highly accurate, rugged and reliable instrument for regulating and measuring the flow of compressed air in a manual powder or wire combustion spray system.

Model: 3AFM (Air Flow Meter)

Flow Unit in both NLPM & SCFM



Gas Flow Meter

These precision wall mounting Flowmeters are recommended to use with Flame Spray Equipment. Ease of reading enables economic operation and rapid fault diagnosis. The unit incorporates separate flowmeter for both Oxygen and Fuel Gas supplies.

Model: 2GFM (Gas Flow Meter)
Supply with % scale of 0 - 100



Wire Control Unit

The Wire Reel Stand with Straightner is specifically designed to dispense wires for flame spraying. A wire straightener is included and the unit is capable of handling either reeled or loose coiled wire or spool.



A Trolley with Reel Stand is also available to move the complete system in shop easily.

OPTIONAL ACCESSORIES

Extension Nozzles

Where deep bores or recesses are to be sprayed, a range of extension nozzles are available. These allow access of the spray at depths up to 1.1 Mtr., whilst straight or deflector air caps can be selected to direct the spray stream to the area required.

Typical use include spraying in internal diameter or hard to reach coating surfaces. For ID coatings, a minimum ID measurement of approximately 8 inches is required to achieve optimal coating quality.

Installation is simple by removing the standard Air Cap Body and Siphon plug and replacing it with the extension. This can be done in about 1-2 minutes without any tools. These are available in various lengths for both Oxy-Acetylene and Oxy-Propane Gun.



Length: 6 INCH / 1 FEET / 2 FEET / 3 FEET

Spray Booth

It is used for controlling sprayed dust and fumes when metallizing operation takes place. It carries away metal dust particles as well as heat. Duct work has adjustable pipe length as well as ball joint to fit individual requirement. Reclaimed materials also have a resale value.



Applications of Various MAKings

- Zinc & Aluminum for anti corrosion coating.
- Ni/Al composite wire for bond coats & self bonding coatings.
- Molybdenum for bond coating.
- Molybdenum for hard bearing applications, excellent resistance to adhesive wear, often used on piston rings, synchromesh cones & journals.
- High-chromium steelflor hard & wear resistance coating.
- Bronze, babbitt for bearing applications.
- Stainless steel, nickel and monel for anti corrosion and wear.
- Al & Ni/Al for wear and oxidation resistance.

Metal Spray Wire

Standard materials wires such as Aluminium, Aluminium Alloy, Tin, Tin Alloy, Nickel, Nickel Alloy, Zinc, Stainless Steel, Brass, Molybdenum, Monel, Copper & many more are available.



SAFETY WEARS

Metallizing Helmet

Light weight & comfortable with excellent vision. Supplied from a separate air supply which should be filtered by Helmet Air Conditioner.



Air Breather

This is installed in between compressed air line originating from fitting to the operators helmets. Activated charcoal is contained in the breather unit which separates the oil traces from the compressed air and clean compressed air is fed to the operator's helmet.



Helmet Air Conditioner

Air from the air breather is fed to the helmet air conditioner and air is contained before it is fed to the helmet for breathing. It has also a regulator knob so that the operator can regulate the temperature of air fed to the helmet. It is self driven and doesn't need external power.



Ear Muff

Light weight & comfortable. Operator & other personnel close to the spray operation must wear. Protect the operator from noise originating during thermal spray.



Protective Glass

Light weight & comfortable with excellent vision. Operator & other personnel close to the spray operation must wear.



Operator Safety Mask

Ensure clean breathing air at all times. Conforms to current legislation on breathing air quality.



Hand Gloves

These are made of rubber and have anti-static properties. Hand gloves protect the operators' hand against flying abrasive and static charge if any.



SPARE PARTS LIST (K6 FLAME JET)

DESCRIPTION	PART NO.	ITEM	DESCRIPTION	PART NO.	ITEM
AIR CAP BODY	K6-1		COUNTER SHAFT PLUG O -RING	K6-14	
SIPHON PLUG-LARGE GAUGE ACETYLENE	K6-2-5		SIPHON PLUG-O-RINGS	K6-14	
SIPHON PLUG-LARGE GAUGE PROPANE/LPG	K6-2-6		VALVE CORE O-RINGS	K6-14	
EA AIR CAP (2)	K6-3-EA		GAS HEAD BODY (Includes K6 40,K6 41)	K6-15	
EC AIR CAP (2)	K6-3-EC		GAS HEAD SUB ASSY.	K6-15A	
AH AIR CAP (2)	K6-3-AH		FRONT WIRE GUIDE 3/16"	K6-23-3/16	
CH AIR CAP (2)	K6-3-CH		FRONT WIRE GUIDE 1/8"	K6-23-1/8	
AB FAN SPRAY AIR CAP (2)	K6-4-AB		REAR WIRE GUIDE ASSEMBLY 3/16" (PLAIN)	K6-24-3/16	
CD FAN SPRAY AIR CAP (2)	K6-4-CD		REAR WIRE GUIDE ASSEMBLY 1/8" (PLAIN)	K6-24-1/8	
GAS HEAD ASSEMBLY	K6-5		SET SCREW FOR REAR WIRE GUIDE	K6-25	
NOZZLE NUT (2)	K6-6		UPPER DRIVE ROLL AXLE SCREW	K6-26	
NOZZLE 3/16 ACETYLENE (2)	K6-7-42		GUN HANDLE	K6-27	
NOZZLE 3/16 PROPANE (2)	K6-7-43		GUN HANDLE SCREW	K6-28	
NOZZLE 1/8 ACETYLENE (2)	K6-7-48		DRIVE ROLL CARRIER PIVOT PIN	K6-30	
NOZZLE 1/8 PROPANE (2)	K6-7-49		GUN HANDLE MOUNTING SCREW	K6-31	
OIL SIGHT PLUG ASSEMBLY (Includes K6-49)	K6-10		BEARING COVER MOUNTING SCREW		
O-RING	K6-13		VALVE HANDLE SCREW		

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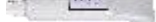
DESCRIPTION	PART NO.	ITEM	DESCRIPTION	PART NO.	ITEM
UPPER DRIVE ROLL 3/16"	K6-32-3/16		OIL SIGHT PLUG O-RING SEAL	K6-49	
UPPER DRIVE ROLL 1/8"	K6-32-1/8		GREASE PLUG SCREW	K6-50	
UPPER DRIVE ROLL GEAR	K6-33		REAR BEARING RETAINER	K6-51	
UPPER DRIVE ROLL SCREW	K6-34		RETAINER PLUG	K6-52	
LOWER DRIVE ROLL 3/16"	K6-35-3/16		TURBINE/GEAR COVER SCREW	K6-53	
LOWER DRIVE ROLL 1/8"	K6-35-1/8		TURBINE ROTOR (Includes K6-67)	K6-54	
LOWER DRIVE GEAR 3/16"	K6-36-3/16		GEAR COVER GASKET	K6-55	
LOWER DRIVE GEAR 1/8"	K6-36-1/8		GEAR COVER	K6-56	
UPPER DRIVE ROLL AXLE	K6-37		PLUG SCREW	K6-57	
BEARING FOLLOWER SUB-ASSEMBLY	K6-39		CENTRAL HOUSING (Includes K6-13, K6-60, K6-61 K6-62 & K6-63)	K6-59	
MOUNTING STUD	K6-40		PUMP HOUSING	K6-60	
MOUNTING STUD ROLL PIN	K6-41		PUMP HOUSING MOUNTING SCREW	K6-61	
VALVE CORE	K6-43		PUMP HOUSING LOCK WASHER	K6-62	
O-RING	K6-44		DOWEL PIN FOR MOUNTING GEAR COVER	K6-63	
VALVE HANDLE ASSEMBLY	K6-45		FINGER SPRING	K6-64	
O-RING	K6-46		DRIVE SHAFT	K6-65	
BEARING COVER	K6-47		DRIVE SHAFT GEAR	K6-66	
BEARING COVER GASKET	K6-48				

SPARE PARTS LIST (K6 FLAME JET)

DESCRIPTION	PART NO.	ITEM
SET SCREW FOR TURBINE ROTOR	K6-67	
GEAR SET SCREW		
SET SCREW		
DRIVE SHAFT SEAL	K6-68	
RIGHT BALL BEARING FOR DRIVE SHAFT	K6-69	
FRONT AND REAR BALL BEARING FOR COUNTER SHAFT	K6-70	
LEFT BALL BEARING FOR DRIVE SHAFT		
NAME PLATE	K6-71	
RIGHT BALL BEARING FOR TURBINE SHAFT	K6-73	
WASHER FOR BRAKE DISC RETURN SPRING	K6-74	
BRAKE DISC RETURN SPRING	K6-76	
TURBINE COVER	K6-77	
BRAKE DISC	K6-78	
O-RING	K6-79	
OIL SEAL BOBBIN	K6-80	
OIL SEAL SLEEVE	K6-81	
OIL SEAL WASHER	K6-82	
LEFT BALL BEARING FOR TURBINE SHAFT	K6-83	

DESCRIPTION	PART NO.	ITEM
NUT FOR COUNTER SHAFT FRONT BEARING	K6-84	
NUT FOR LEFT BEARING		
CONTROL RING	K6-85	
SPIDER ASSEMBLY	K6-86	
DETENT SPRING	K6-87	
SCREW FOR RING NAMEPLATE	K6-89	
COUNTER SHAFT	K6-91	
TURBINE SHAFT, HIGH SPEED	K6-92-H	
TURBINE SHAFT, STANDARD	K6-92-S	
COUNTER SHAFT GEAR HIGH SPEED	K6-93-H	
COUNTER SHAFT GEAR STANDARD	K6-93-S	
GAS HEAD MOUNTING SCREW REAR	K6-96	
GAS HEAD MOUNTING SCREW FRONT	K6-97	
FACE BLOCK	K6-98	
SPRING	K6-99	
SOCKET WRENCH	K6-100	
LIGHTER	K6-101	
TURBINE SHAFT BEARING NUT-RIGHT	K6-102	

SPARE PARTS LIST (K6 FLAME JET)

DESCRIPTION	PART NO.	ITEM
DRIVE ROLL TOOL	K6-103	
FIXING TOOL	K6-104	
SCREW DRIVER	K6-105	
FIX SPANNER (19/22)	K6-106	
NOZZLE O-RING, LARGE	K6-108	
NOZZLE O-RING, SMALL	K6-109	
GEAR LUBE	K6-110	
VALVE LUBE	K6-111	
SPANNER (13 x 17)		
CLEANING WIRE KIT	K6-114	
SUPPORT BLOCK MOUNTING SCREW	K6-117	
SNUBBER ASSEMBLY 1/8"	K6-118-1/8	
SNUBBER ASSEMBLY 3/16"	K6-118-3/16	
ROLLER SPRING	K6-119	
PUSH ROD	K6-123	
TOOL POST FIXTURE	K6-124	
CLAMP SHAFT HOUSING	K6-126	
CLAMP SHAFT	K6-127	

DESCRIPTION	PART NO.	ITEM
PIN	K6-128	
PUSH ROD SEAL	K6-129	
HANDLE ASSEMBLY	K6-130	
LOCKING SCREW	K6-131	
SPRING CAP	K6-132	
SET SCREW	K6-133	
CAM FOLLOWER ASSEMBLY	K6-134	
CAM ASSEMBLY	K6-135	
DRIVE ROLL CARRIER	K6-136	
SUPPORT BLOCK SUB ASSY.	K6-139	
SUPPORT BLOCK COMPLETE	K6-139A	
REAR WIRE GUIDE 3/16 (Plain)	K6-140	
REAR WIRE GUIDE 1/8 (Plain)	K6-141	
STRAIN RELIEF ASSEMBLY	K6-143	
STRAIN RELIEF BUSHING	K6-144	
COVER MOUNTING SCREW	K6-149	
FIXTURE BAR	K6-150	
FACE BLOCK MFG. SCREW	K6-156	

SPARE PARTS LIST (K6 FLAME JET)

DESCRIPTION	PART NO.	ITEM	DESCRIPTION	PART NO.	ITEM
HOSE CONNECTION BLOCK SCREW	K6-157		GUIDE INSERT 1/8	K6-187 1/8	
OIL SEAL ASSEMBLY (Includes K6-79, K6-80, K6-81, K6-82 & K6-283)	K6-160		SNUBBING REAR GUIDE 3/16	K6-188 3/16	
SPRING (Strain Relief Spring)	K6-163		SNUBBING REAR GUIDE 1/8	K6-188 1/8	
PLUNGER(Fuel, Gas & Oxygen)	K6-165		SIPHON PLUG AND NOZZLE O-RING PACKAGE (Includes K6-14, K6-108, K6-109)	K6-190	
PLUNGER(Air)	K6-166		SIPHON PLUG ASSEMBLY 1/8, PROPANE	K6-200-2	
PLUNGER O-Ring (Fuel, Gas & Oxygen)	K6-168		SIPHON PLUG ASSEMBLY 1/8, ACETYLENE	K6-200-3	
PLUNGER O-Ring(Air)	K6-169		SIPHON PLUG ASSEMBLY 3/16, PROPANE	K6-200-4	
PLUNGER SEAL O-RING (Gas, Fuel & Oxygen)	K6-170		SIPHON PLUG ASSEMBLY 3/16, ACETYLENE	K6-200-5	
PLUNGER SPRING	K6-171		HEX ALLEN KEY 2mm	K6-202	
DETENT PIN	K6-173		HEX ALLEN KEY 3mm	K6-240	
DETENT PIN SPRING	K6-174		SPRING RETAINER	K6-241	
HANDLE WASHER	K6-175		ROLLER	K6-242	
HOSE CONNECTION BLOCK	K6-176		SNUBBER SPRING HOUSING	K6-243	
VALVE CORE STOP SCREW	K6-179		SPRING	K6-245	
GAS HEAD VALVE O-RING PACKAGE(CompletesetofO-Ring for K6-5 Gas Head)	K6-184		COMPLETE O RING KIT	K6-262	
PLUG SCREW O-RING	K6-186		OIL SEAL SPRING	K6-283	
PLUNGER O-RING (AIR)			The MAK K6 is a direct replacement for Metco 12E Gun. All the parts in this gun are interchangeable with Metco parts. The part number suffix (i.e the portion after the dash "-") should match the Metco part number. For example: If you wish to order a K6-7-48 Nozzle, the MAK Part No. would be K6-7-48		
GUIDE INSERT 3/16	K6-187 3/16				



MAK ENGINEERS

H.O. & Works:



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